Sistemas conexionistas*

Senén Barro Ameneiro, CiTIUS

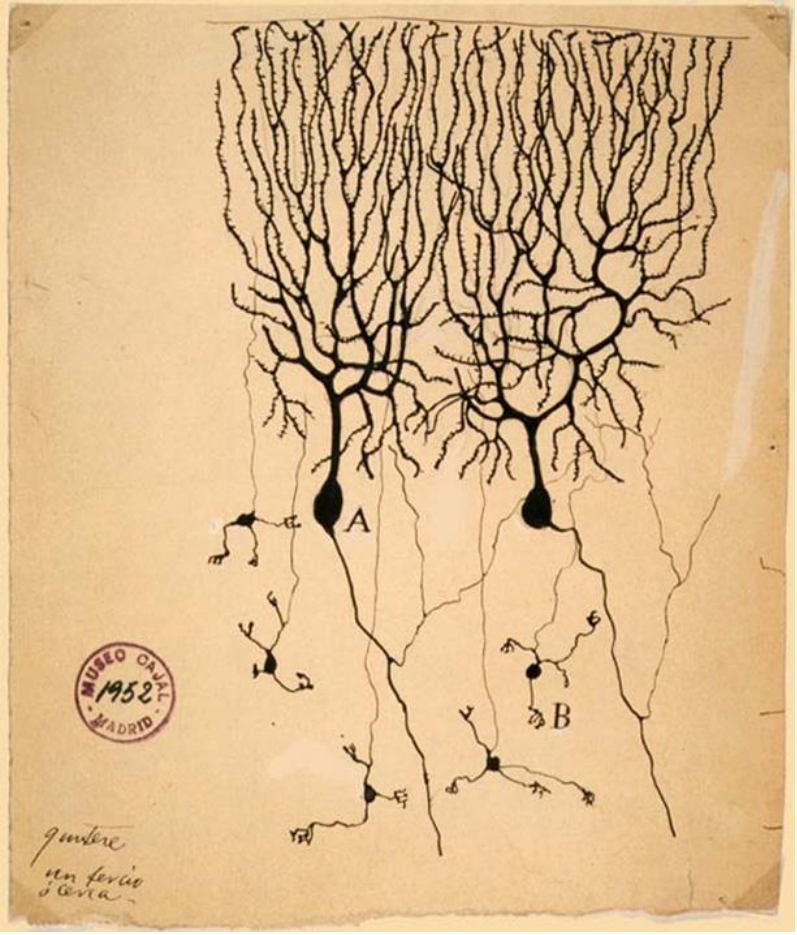
@SenenBarro

*Baseado parcialmente en:
http://elvex.ugr.es/courses.html

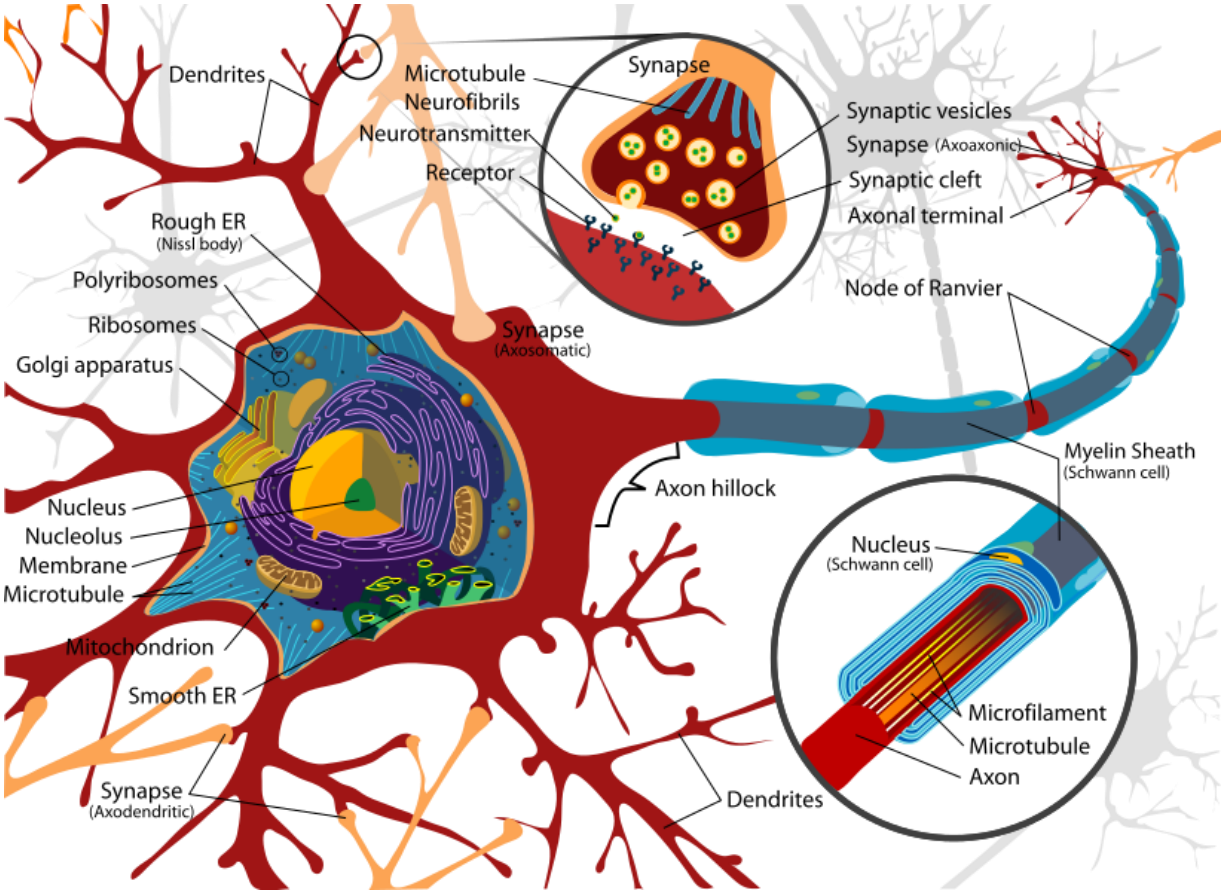
Sistemas conexionistas

Nos sistemas conexionistas ou Redes Neuronais Artificiais (RNA) apréciase especialmente a dualidade científica e tecnolóxica da IA:

- Bioinspiración
- Tecnoinspiración

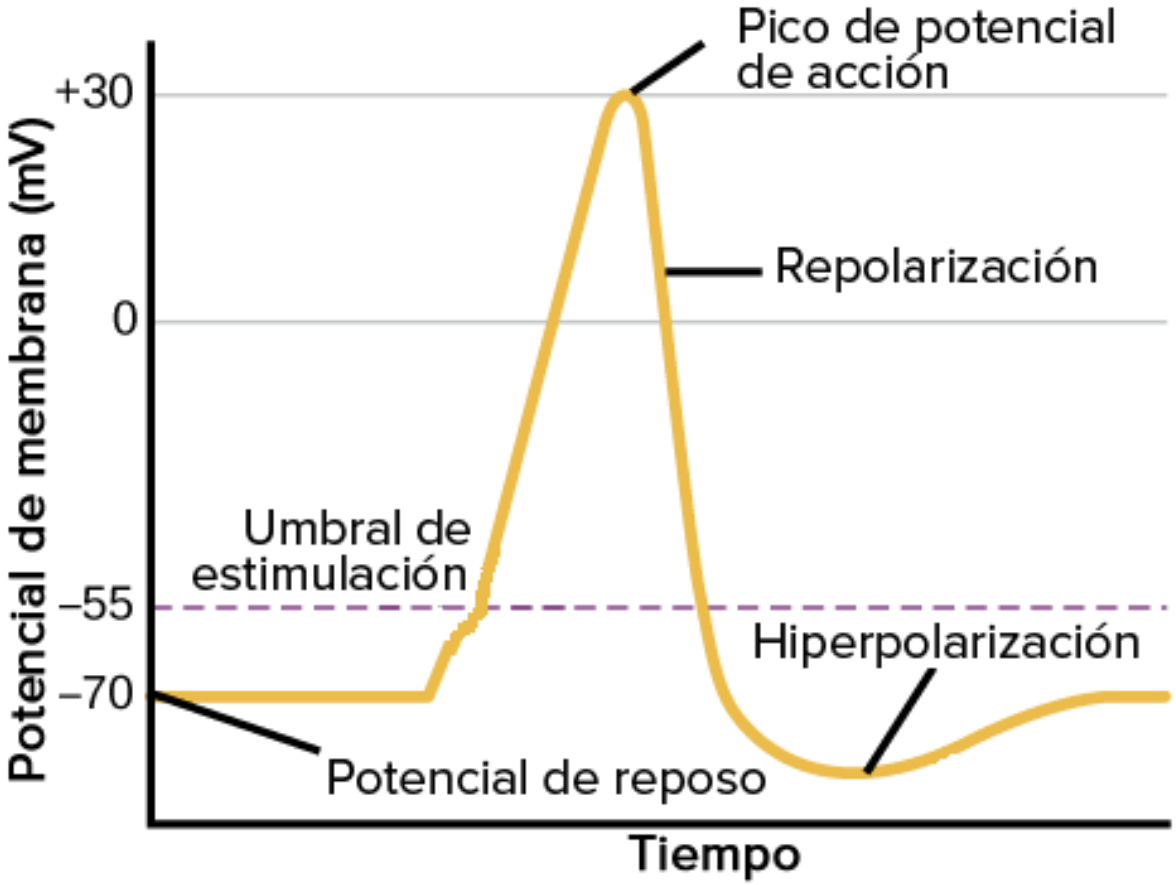


Sistemas conexionistas



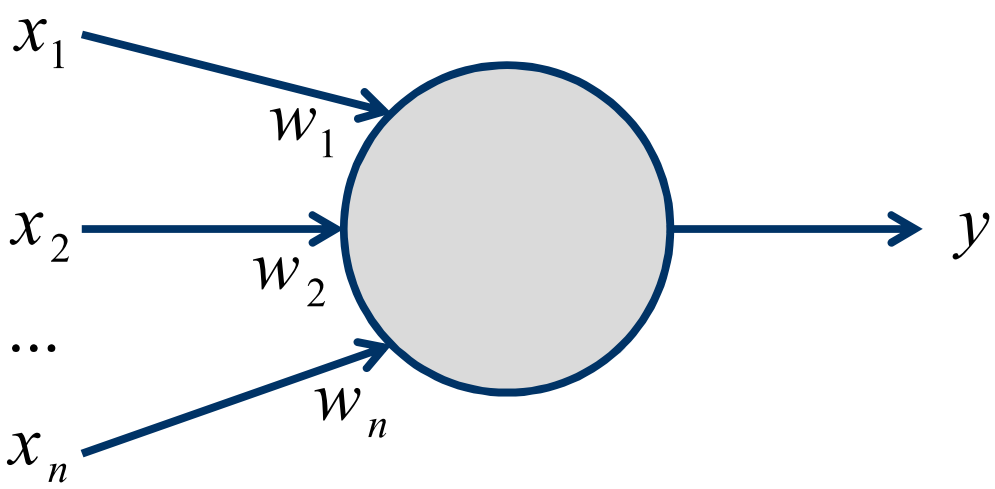
Neurona (https://es.wikipedia.org/wiki/Neurona)

Sistemas conexionistas



Potencial de acción (ideal) dunha neurona

Sistemas conexionistas



$$y = \sum_i x_i w_i = x_1 w_1 + x_2 w_2 + \ldots + x_n w_n$$

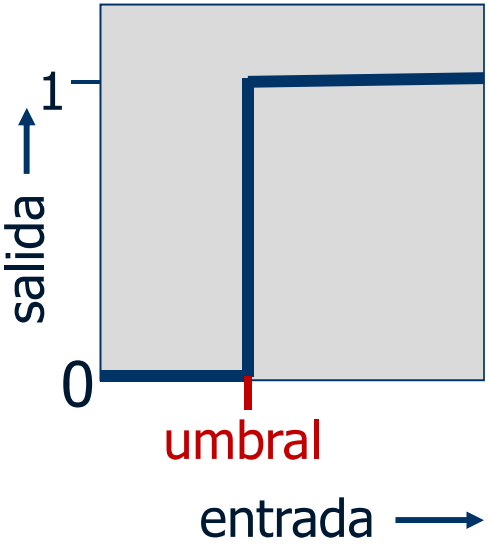
Modelo máis simple de neurona artificial

Sistemas conexionistas

[McCulloch & Pitts, 1943]

$$z = b + \sum_i x_i w_i$$

$$y = \begin{cases} 1 & si\ z \geq 0 \\ 0 & en\ otro\ caso \end{cases}$$



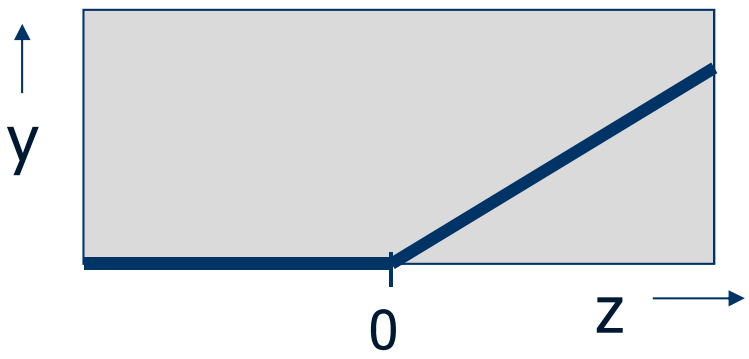
Umbral θ=-b

Neurona binaria con umbral

Sistemas conexionistas

$$z = \sum_i x_i w_i$$

$$y = \begin{cases} z & si\ z \geq 0 \\ 0 & en\ otro\ caso \end{cases}$$



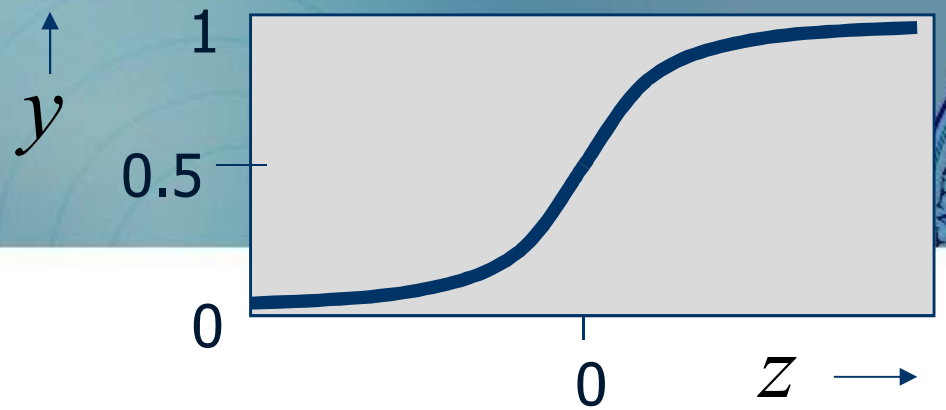
Asumiendo x_0=1 y w_0=b (umbral θ=-b)

Neurona lineal rectificada

Sistemas conexionistas

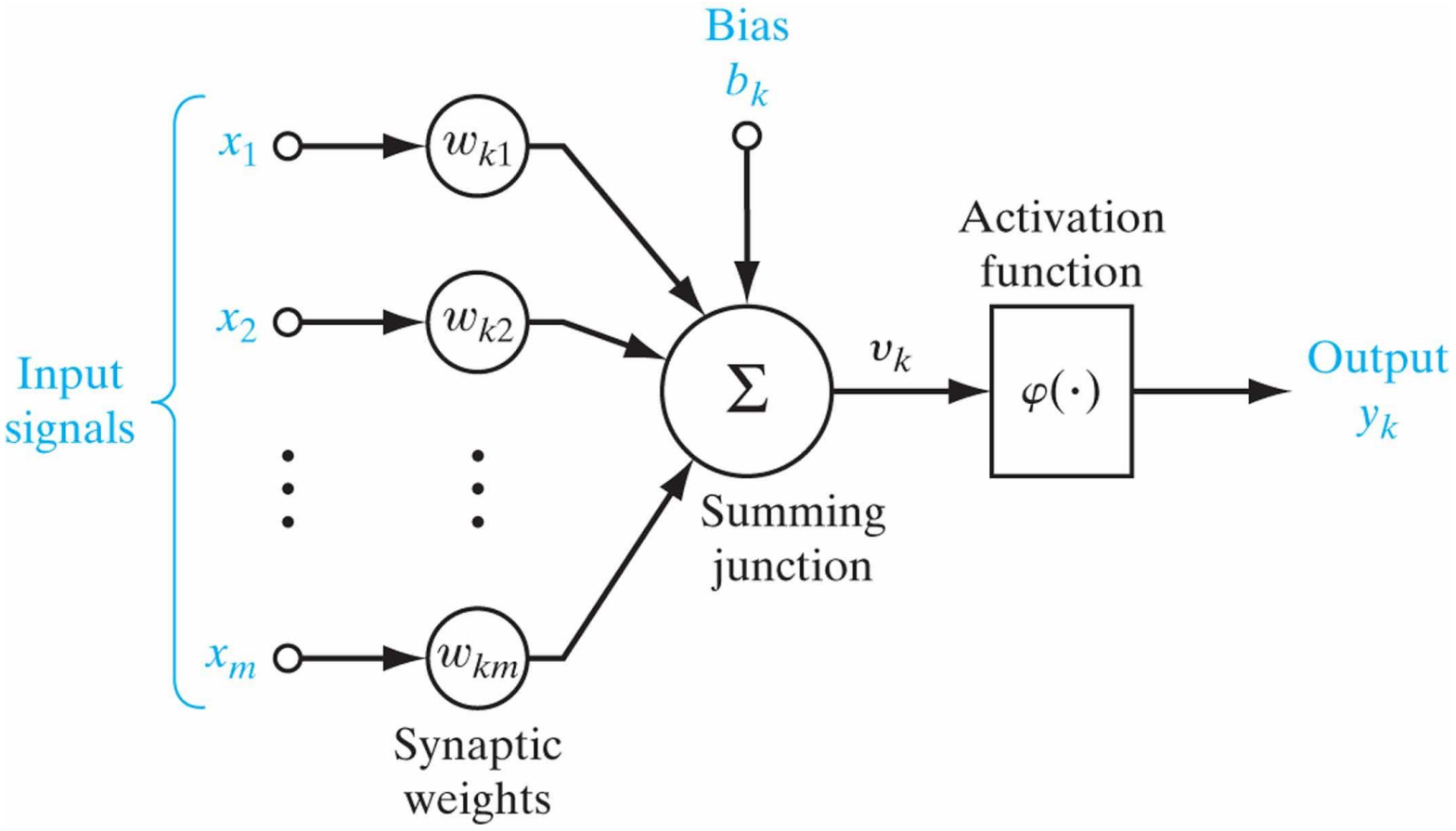
$$z = \sum_i x_i w_i$$

$$y = \frac{1}{1+e^{-z}}$$



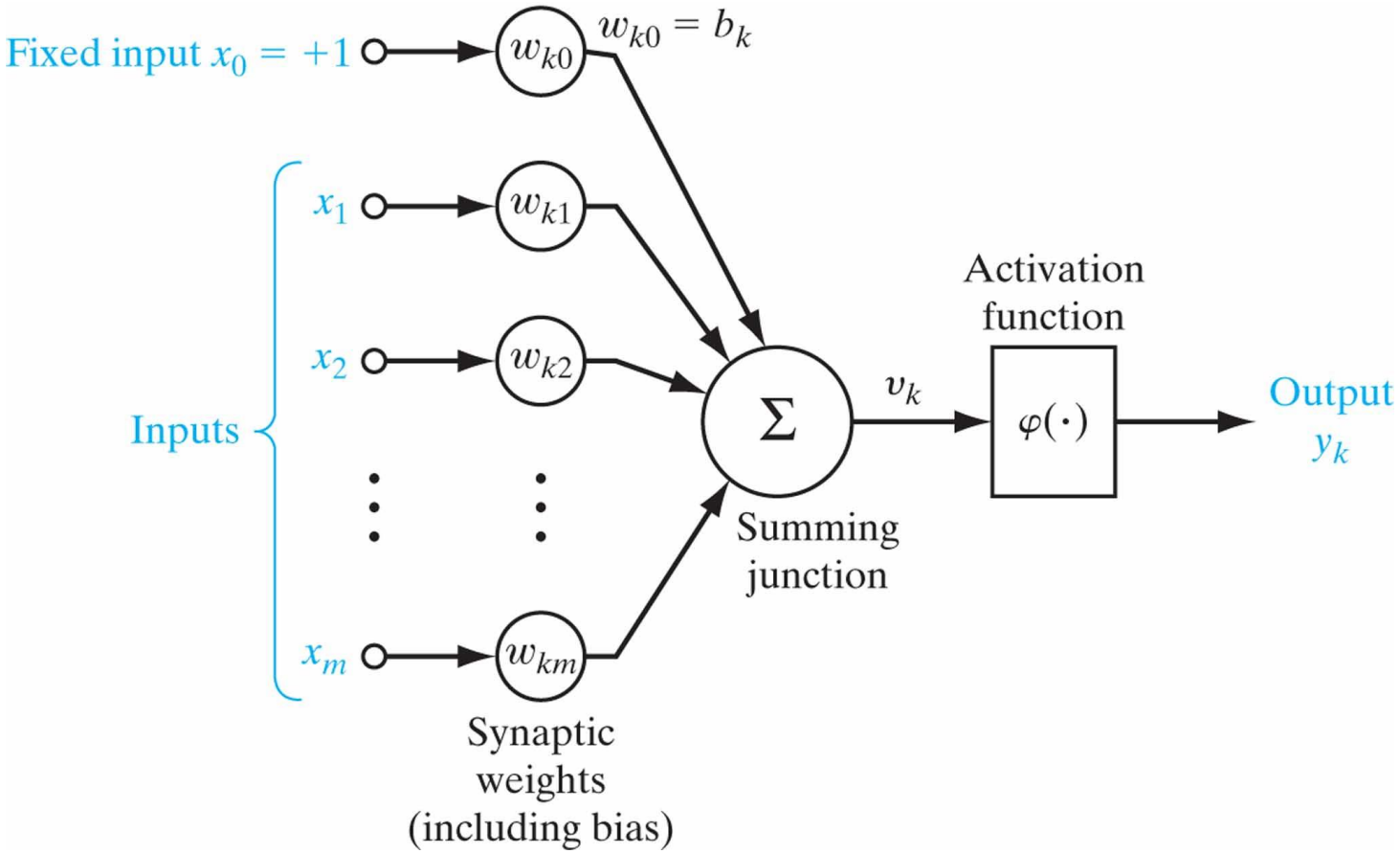
Neurona *sigmoidal*

Sistemas conexionistas



Representación común de una Neurona Artificial

Sistemas conexionistas

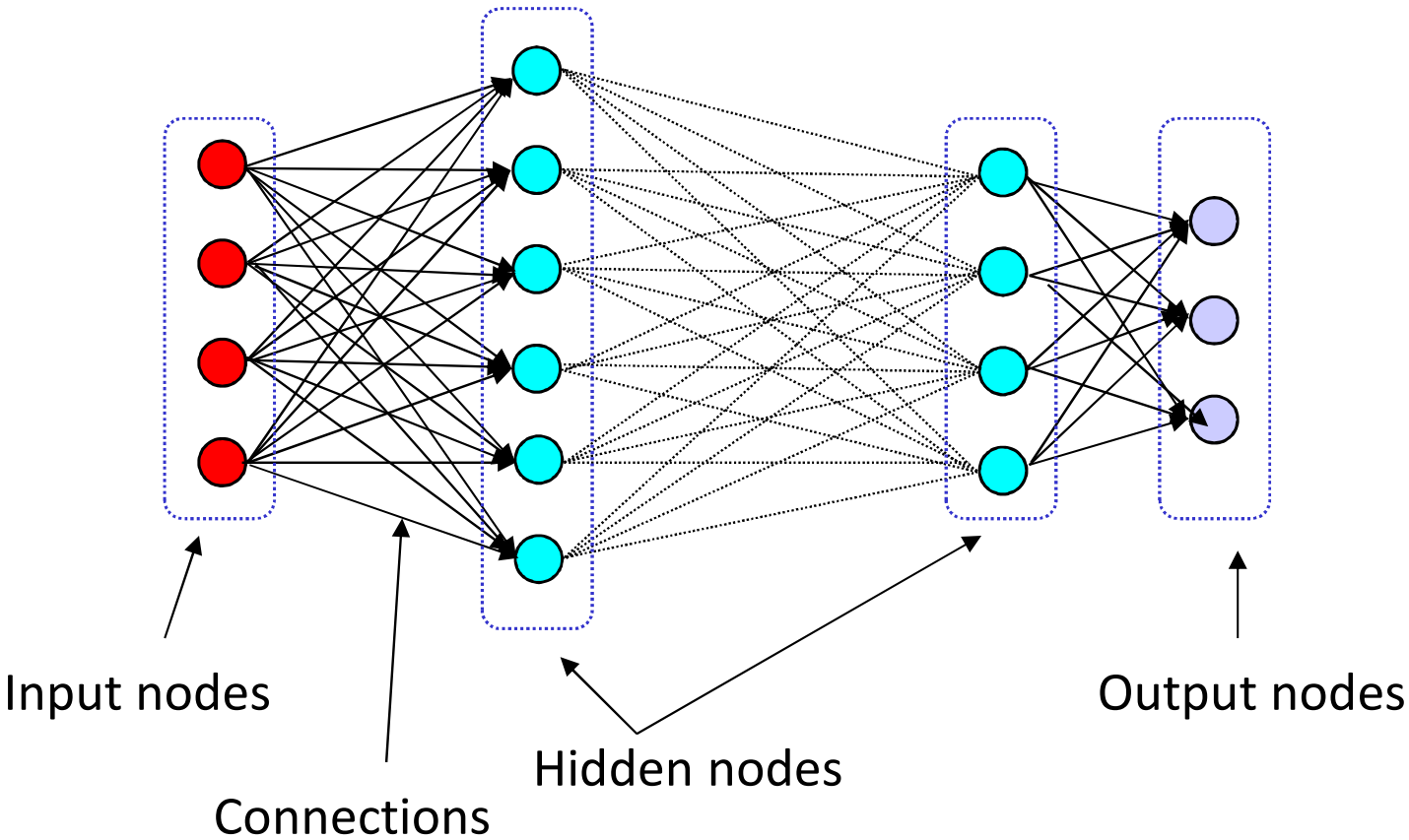


Neurona Artificial con *bias* integrado como un peso

Sistemas conexionistas

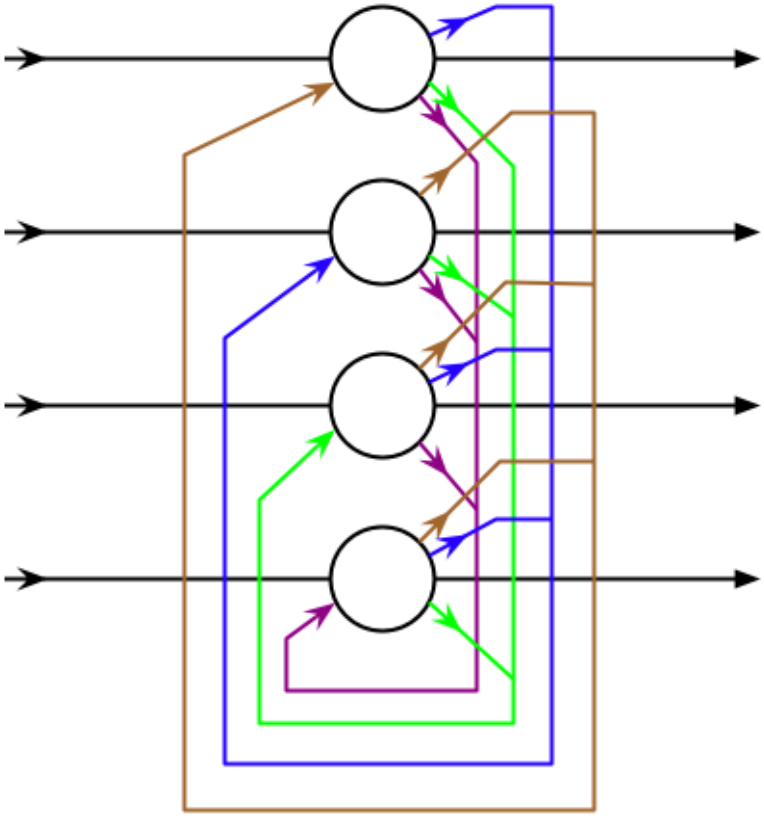
- 1943 Neurona formal de McCulloch e Pitts
- 1957 Perceptrón, Frank Rosenblatt
- 1969 Publicación do libro “Perceptrons”, Minsky e Papert
- 1974-1986 Retropropagación (backpropagation)
- 1982 Redes de Hopfield
- 2006 Aprendizaxe profunda (*Deep Learning*)

Sistemas conexionistas



Rede Neuronal Multicapa

Sistemas conexionistas

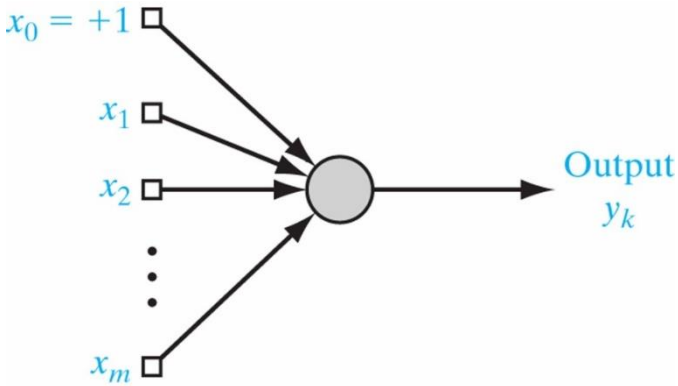


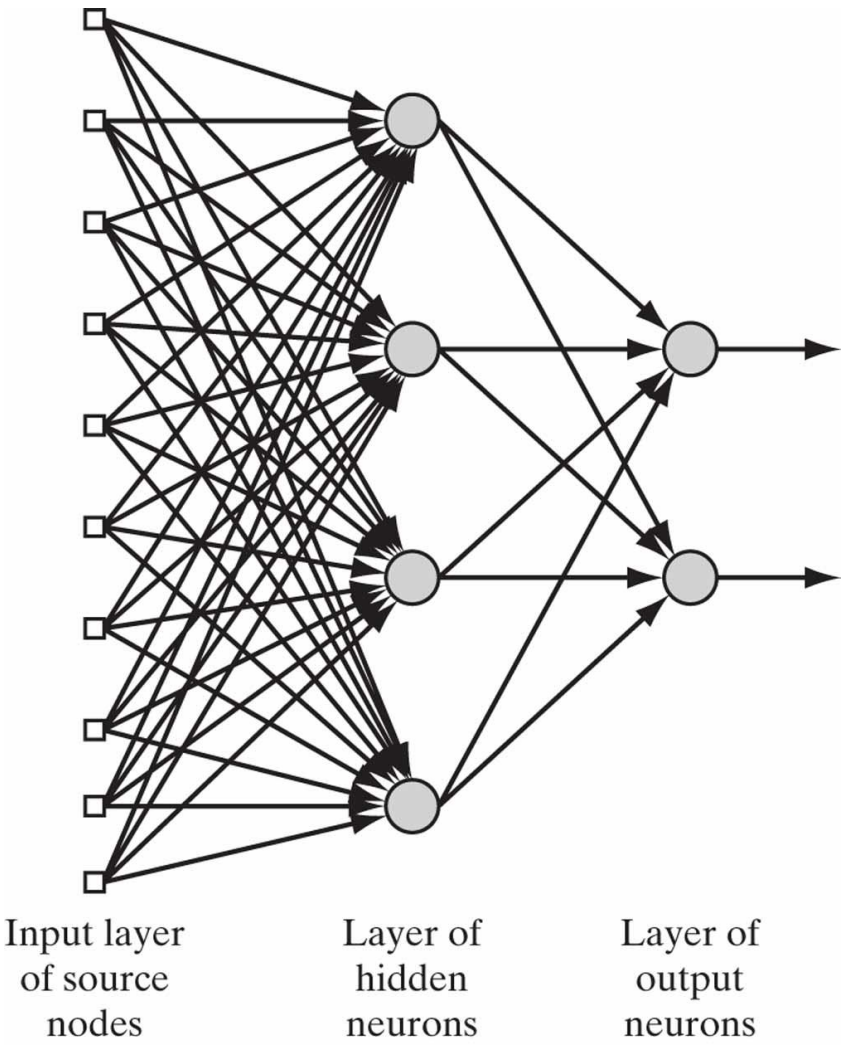
Rede de Hopfield

Sistemas conexionistas

- Redes Neuronais Artificiais (RNA)
  - **Estrutura** composta por Neuronas Artificiais (NA), profusamente conectadas entre si
  - Diferentes **modelos** ou arquitecturas dependendo do tipo de NA e da súa organización
  - Arquitecturas de **cómputo** paralelo e distribuido
  - **Aprendizaxe** a partir de datos de entrada e, en xeral, mediante modificación de pesos sinápticos

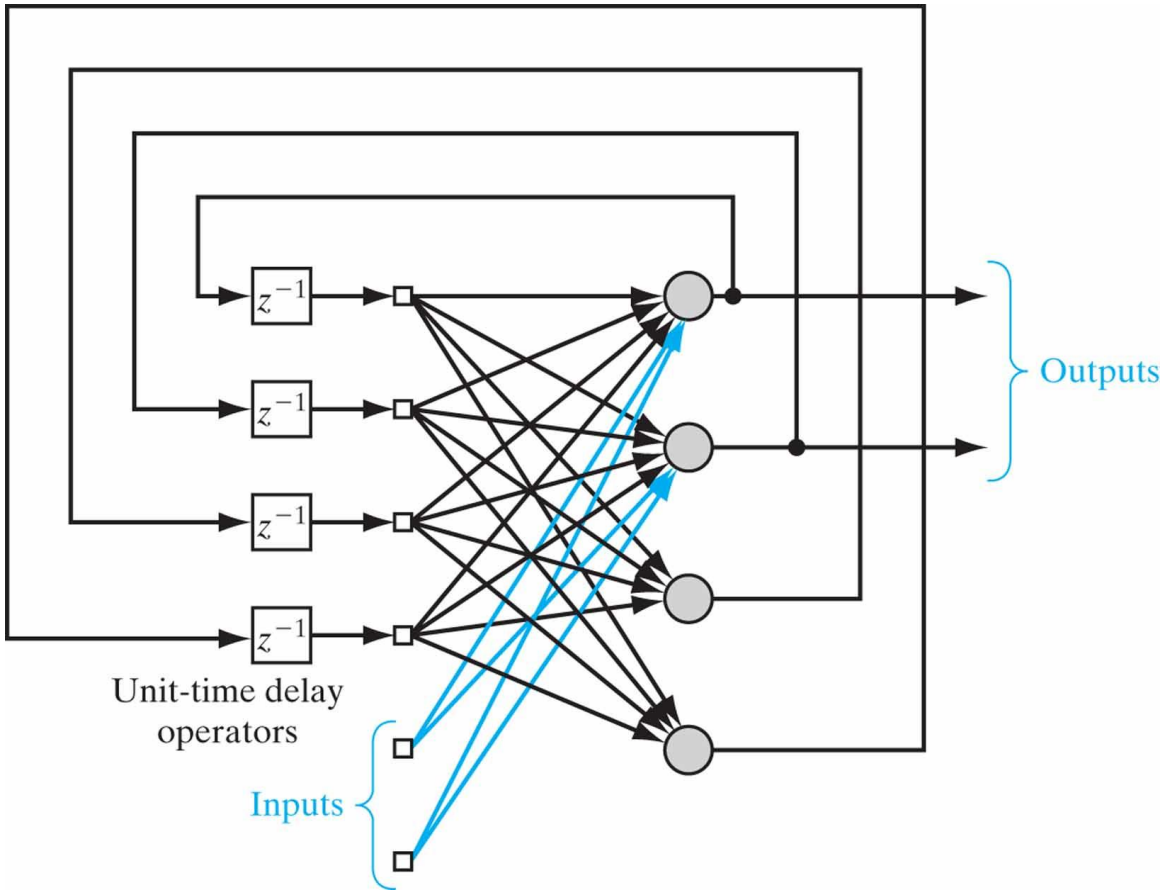
Sistemas conexionistas





Perceptrón e arquitectura multicapa cara adiante

Sistemas conexionistas



Rede recurrente

Sistemas conexionistas

- Como aprende unha RNA con *Perceptróns*?
  - **Aprendizaxe supervisada** a partir de exemplos –conxunto de entrenamento-
  - **Conxunto de entrenamento** formado por pares (entrada, saída desexada), suficientemente representativo do problema a resolver
  - **Apréndese axustando os pesos** das entradas dos Perceptróns para que no posible a Rede responda coa saída desexada ante cada exemplo do conxunto de entrenamento

Sistemas conexionistas

TABLE 1.1 Summary of the Perceptron Convergence Algorithm

Variables and Parameters:

$$
\begin{aligned}
\mathbf{x}(n) &= (m+1)\text{-by-1 input vector} \\
&= [+1, x_1(n), x_2(n), \ldots, x_m(n)]^T \\
\mathbf{w}(n) &= (m+1)\text{-by-1 weight vector} \\
&= [b, w_1(n), w_2(n), \ldots, w_m(n)]^T \\
b &= \text{bias} \\
y(n) &= \text{actual response (quantized)} \\
d(n) &= \text{desired response} \\
\eta &= \text{learning-rate parameter, a positive constant less than unity}
\end{aligned}
$$

1. *Initialization.* Set $\mathbf{w}(0) = \mathbf{0}$. Then perform the following computations for time-step $n = 1, 2, \ldots$.
2. *Activation.* At time-step n, activate the perceptron by applying continuous-valued input vector $\mathbf{x}(n)$ and desired response $d(n)$.
3. *Computation of Actual Response.* Compute the actual response of the perceptron as

$$y(n) = \operatorname{sgn}[\mathbf{w}^T(n)\mathbf{x}(n)]$$

   where $\operatorname{sgn}(\cdot)$ is the signum function.
4. *Adaptation of Weight Vector.* Update the weight vector of the perceptron to obtain

$$\mathbf{w}(n+1) = \mathbf{w}(n) + \eta[d(n) - y(n)]\mathbf{x}(n)$$

   where

$$d(n) = \begin{cases} +1 & \text{if } \mathbf{x}(n) \text{ belongs to class } \mathscr{C}_1 \\ -1 & \text{if } \mathbf{x}(n) \text{ belongs to class } \mathscr{C}_2 \end{cases}$$

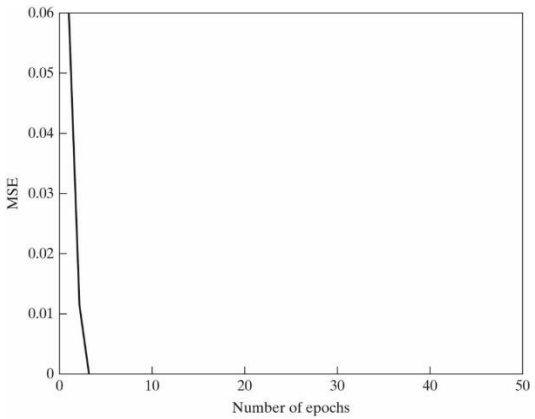
5. *Continuation.* Increment time step n by one and go back to step 2.

Algoritmo de converxencia do Perceptrón

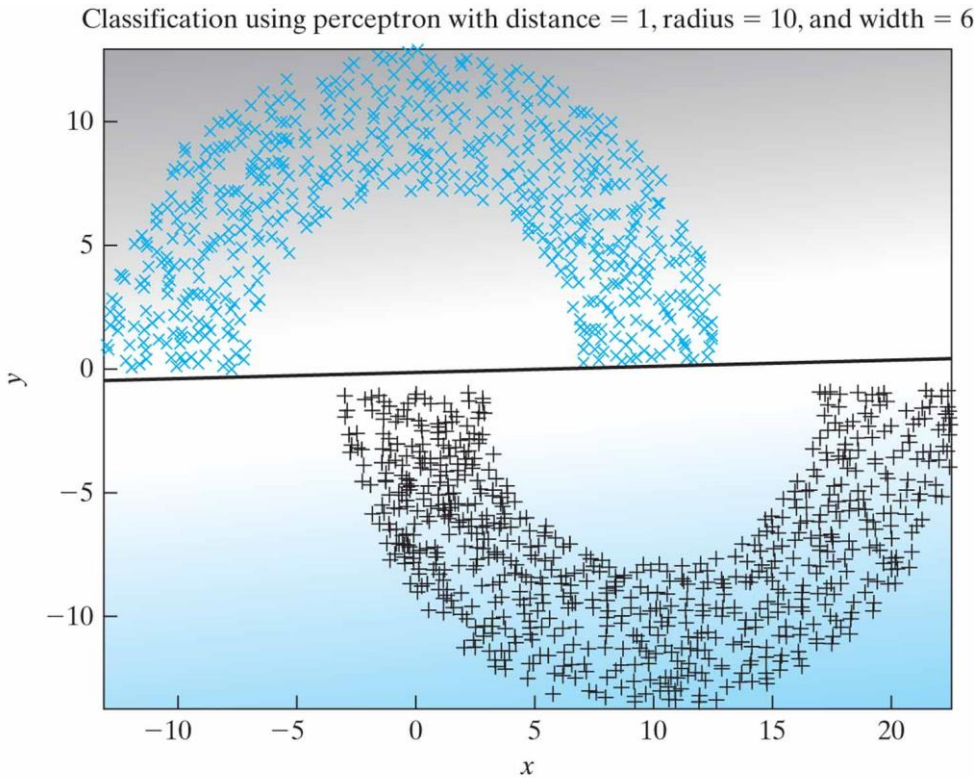
Sistemas conexionistas

- **Algoritmo de converxencia** do *Perceptrón*
  - **Atopa unha solución se esta existe** -conxunto de pesos que clasifican correctamente o conxunto de entrenamento-
  - Só pode distinguir **categorías linealmente separables**
  - Nunha **arquitectura multicapa** pode discriminar categorías non linealmente separables, pero necesita outro tipo de algoritmos de aprendizaxe

Perceptrón



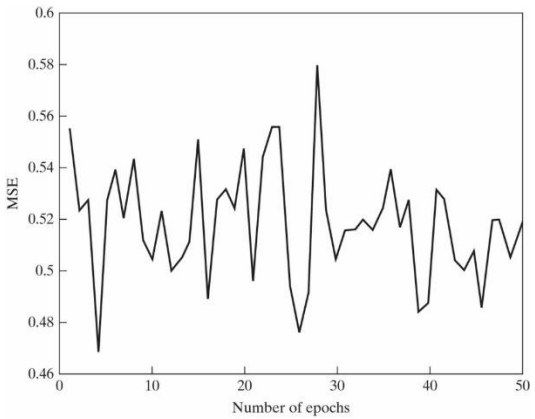
(a) learning curve



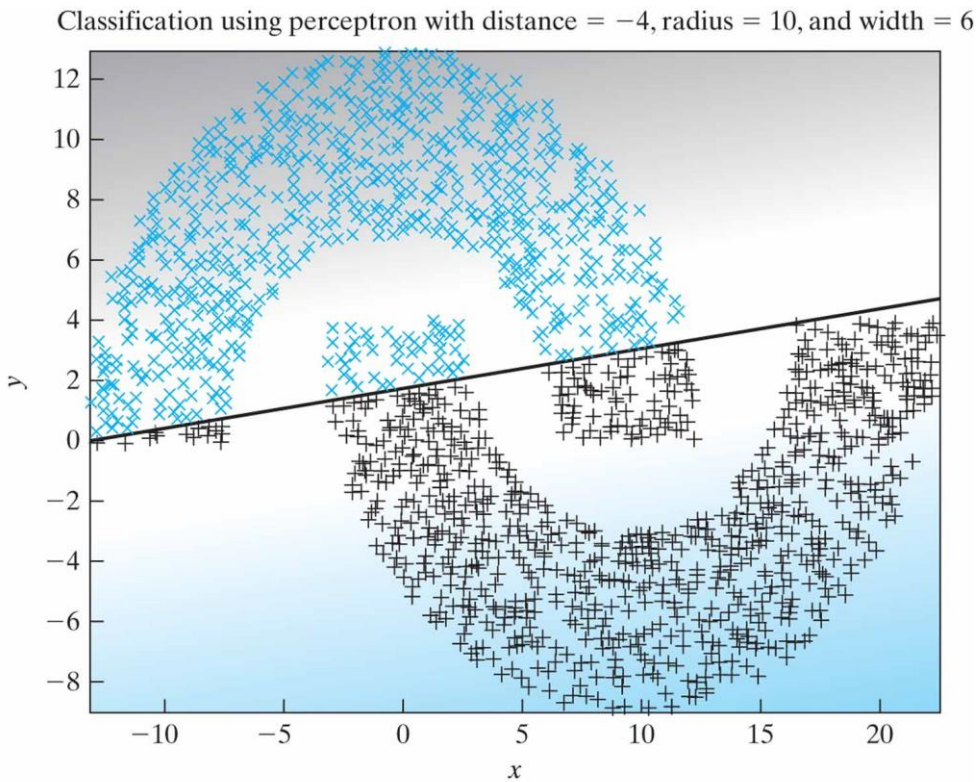
(b) testing result

Clases linealmente separables

Perceptrón



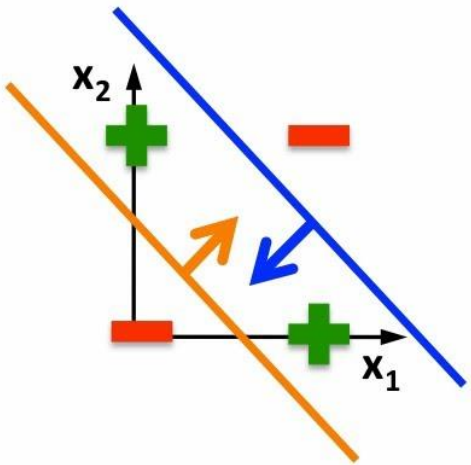
(a) learning curve

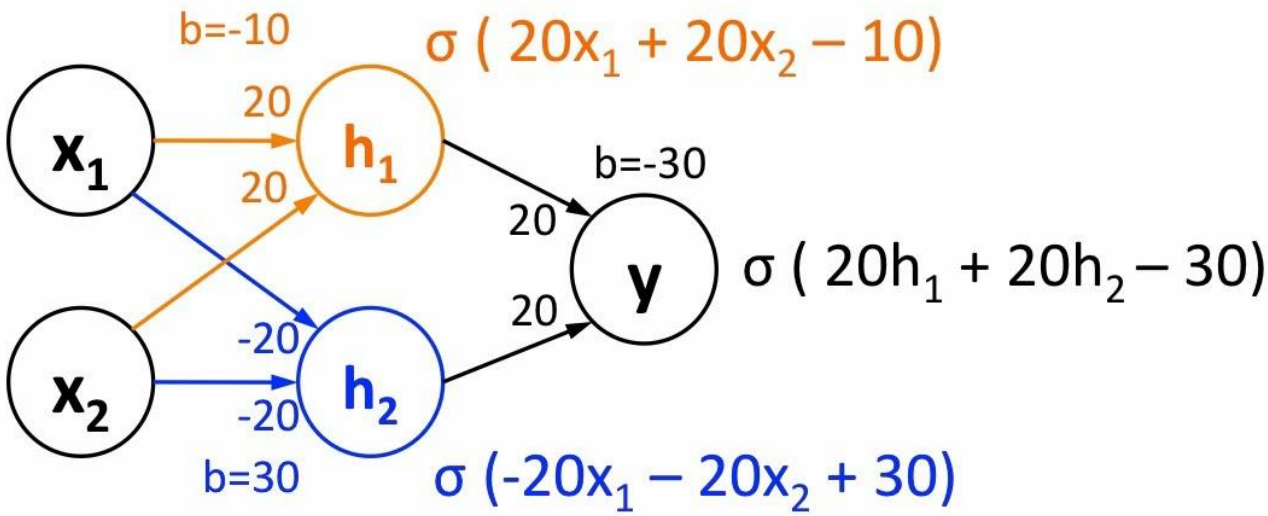


(b) testing result

Clases non linealmente separables

Linear classifiers cannot solve this





$\sigma(20*0 + 20*0 - 10) \approx 0$ $\quad$ $\sigma(-20*0 - 20*0 + 30) \approx 1$ $\quad$ $\sigma(20*0 + 20*1 - 30) \approx 0$

$\sigma(20*1 + 20*1 - 10) \approx 1$ $\quad$ $\sigma(-20*1 - 20*1 + 30) \approx 0$ $\quad$ $\sigma(20*1 + 20*0 - 30) \approx 0$

$\sigma(20*0 + 20*1 - 10) \approx 1$ $\quad$ $\sigma(-20*0 - 20*1 + 30) \approx 1$ $\quad$ $\sigma(20*1 + 20*1 - 30) \approx 1$

$\sigma(20*1 + 20*0 - 10) \approx 1$ $\quad$ $\sigma(-20*1 - 20*0 + 30) \approx 1$ $\quad$ $\sigma(20*1 + 20*1 - 30) \approx 1$

Copyright © 2014 Victor Lavrenko

Perceptrón

Función XOR